Reg. No. | | | | | | | | | | |

[12CE73]

SEVENTH SEMESTER B.E EXAMINATION DECEMBER - 2015

PRE STRESSED CONCRETE

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Assume suitably any missing data.

1. a. With a neat sketch, explain stress distribution at a section in beam due to dead load, live load and Pre-stress. **08**

b. Write a note on materials used in prestressed concrete. **06**

2. a. A concrete beam having a rectangular section 150 x 300 is prestressed by a parabolic cable at an eccentricity of 75mm at mid-span towards bottom (soffit) and at eccentricity of 25mm towards top at support sections. The effective pre-stressing force is 350KN. The beam supports a concentrated load of 20KN at center of span in addition to the self-weight with a span at 8m. Find the short term deflection at centre of span under prestress, self-weight and live load. Find also the long term deflection if the loss ratio is 0.8 and the creep coefficient is 1.6. Take $E_c = 38KN/mm^2$. **08**

b. Explain the effect of tendon profile on deflection. **06**

3. a. A prestressed concrete beam, 200mm x 300mm is prestressed with wires (area=320mm^2) located at a constant eccentricity of 50mm and carrying an initial stress of 1000N/mm^2. The span of the load beam is 10m. Calculate the percentage loss of stress in wires if **10**

i. The beam is pre-tensioned

ii. The beam is post-tensioned, using the following data:

$$E_s = 210KN/mm^2 \quad and \quad E_c = 35KN/mm^2$$

Relaxation of steel stress = 5 Percent of the initial stress shrinkage of Concrete = 300 x 10^{-6} for pretensioning and 200 x 10^{-6} for post tensioning.

Creep coefficient = 1.6. slip at anchorage = 1mm.

Frictional coefficient for wave effect = 0.0015/m

b. Write a note on classification of losses of prestress. **04**

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4. a. A post tensioned bridge girder with unbonded tendons is of box section of overall dimensions 1200mm wide by 1800mm deep, with wall thickness of 150mm. The high-tensile steel has an area of $4000mm^2$ and is located at an effective depth of 1600mm. The effective prestress in steel after all losses is 1000 N/mm^2 and the effective span of girder is 24m. If f_{ck} = $40N/mm^2$ and f_p=$1600N/mm^2$., estimate the ultimate flexural strength of the section. Use IS-1343:1980 method. **07**

b. A pretensioned, prestressed concrete beam having a rectangular section, 150mm wide and 350mm deep has an effective cover of 50mm. If f_{ck} = $40N/mm^2$ and f_p=$1600N/mm^2$, and the area of prestressing steel A_p=$461mm^2$, calculate the ultimate flexural strength of the section using strain compatibility method. **07**

5. a. A fieyssinet, anchorage 100mm dia carrying 12 wires of 7mm dia is embedded concentrically in the web of an I-Section of Web thickenss 225mm. Using IS Code method, determine the tensile and bursting tensile force in the end block. Design the end block and sketch the reinforcement details. **07**

b. Write a note on classification of members based on prestressing force or amount of prestressing. **07**

6. A partially prestressed, pre-tensioned mast is to be designed to suit the following data: **14**

- Spacing of poles = 100m
- Free-standing height of the pole above ground = 10m
- The pole is to carry twin-conductor high-voltage lines (60cm apart) on a cross tree at 9m above ground level. Conductor size: effective overall diameter=10mm.
- Tension in each conductor = 5KN.
- Poles are to be located in zone where wind pressure is $10N/mm^2$
- 28-day cube strength of concrete = $50N/mm^2$.
- Modules of elasticity of concrete = $40.5KN/mm^2$.
- Modulus of rupture of concrete = $5N/mm^2$.
- High tensile wires of 5mm dia available.
- Ultimate tensile strength = 1600 N/mm^2.
- Loss ratio = 0.7
- Permissible compressive stress in concrete f_{cw}=18 N/mm^2.
- Permissible Tensile stress in concrete, f_{tw} = 5 N/mm^2.

7. a. A composite T girder of span 5m is made up of a pre-tensioned rib, 100mm wide by 200mm deep, with an in situ cast slab, 400mm wide and 40mm thick The rib is prestressed by a straight cable having an eccentricity of 33.33mm and carrying an initial force of 150KN. The loss of prestress may be assumed to be 15 percent check the composite T-beam for the limit state of deflection if it supports on imposed load of 3.2 KN/m for **08**

i. Unpropped construction

ii. Propped construction

Assume a modulus of elasticity of 35KN mm^2 for both precast and in Situ

b. The cross section of a composite beam is of T-Section having pretensioned rib 80mm wide and 240mm deep, and an in Situ cast slab, 350mm wide and 80mm thick. The pre-tensioned beam is reinforced with eight wires of 5mm diameter with an ultimate tensile strength of $1600N/mm^2$, located 60mm from the soffit of the beam. The compressive strength of concrete in the in Situ cast and precast elements is 20 and $40N/mm^2$ respectively. If adequate reinforcement are provided to prevent shear failure at the interface, estimate the flexural strength of the composite section. **06**

8. a. Write a note on advantages of prestressed concrete bridges. **06**

b. Explain in detail, the principles of design of prestressed concrete bridges. **08**